DAA ASSIGNMENT

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RED – BLACK TREE

CODE: -

```
  GNU nano 7.2                                                    redblack.c
#include <stdio.h>
#include <stdlib.h>

#define RED 1
#define BLACK 0

struct Node {
    int data;
    int color;
    struct Node *left, *right, *parent;
};

struct Node *root = NULL;

struct Node* createNode(int data) {
    struct Node *newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = data;
    newNode->color = RED;
    newNode->left = newNode->right = newNode->parent = NULL;
    return newNode;
}

void rotateLeft(struct Node *x) {
    struct Node *y = x->right;
    x->right = y->left;
```

```
    if (y->left != NULL)
        y->left->parent = x;

    y->parent = x->parent;

    if (x->parent == NULL)
        root = y;
    else if (x == x->parent->left)
        x->parent->left = y;
    else
        x->parent->right = y;

    y->left = x;
    x->parent = y;
}

void rotateRight(struct Node *x) {
    struct Node *y = x->left;
    x->left = y->right;

    if (y->right != NULL)
        y->right->parent = x;

    y->parent = x->parent;
```

```
    if (x->parent == NULL)
        root = y;
    else if (x == x->parent->right)
        x->parent->right = y;
    else
        x->parent->left = y;

    y->right = x;
    x->parent = y;
}

void fixViolation(struct Node *z) {
    while (z != root && z->parent->color == RED) {
        struct Node *parent = z->parent;
        struct Node *grandparent = parent->parent;

        if (parent == grandparent->left) {
            struct Node *uncle = grandparent->right;

            if (uncle && uncle->color == RED) {
                grandparent->color = RED;
                parent->color = BLACK;
                uncle->color = BLACK;
                z = grandparent;
```

```
            } else {
                if (z == parent->right) {
                    z = parent;
                    rotateLeft(z);
                }

                parent->color = BLACK;
                grandparent->color = RED;
                rotateRight(grandparent);
            }
        } else {
            struct Node *uncle = grandparent->left;

            if (uncle && uncle->color == RED) {
                grandparent->color = RED;
                parent->color = BLACK;
                uncle->color = BLACK;
                z = grandparent;
            } else {
                if (z == parent->left) {
                    z = parent;
                    rotateRight(z);
                }

                parent->color = BLACK;
```

```
                parent->color = BLACK;
                grandparent->color = RED;
                rotateLeft(grandparent);
            }
        }
    }

    root->color = BLACK;
}

void insert(int data) {
    struct Node *z = createNode(data);
    struct Node *y = NULL;
    struct Node *x = root;

    while (x != NULL) {
        y = x;
        if (z->data < x->data)
            x = x->left;
        else
            x = x->right;
    }

    z->parent = y;
```

```c
    if (y == NULL)
        root = z;
    else if (z->data < y->data)
        y->left = z;
    else
        y->right = z;

    fixViolation(z);
}

void inorder(struct Node *root) {
    if (root == NULL)
        return;

    inorder(root->left);
    printf("%d(%s) ", root->data, root->color == RED ? "R" : "B");
    inorder(root->right);
}

int main() {
    int n, val;

    printf("Enter number of nodes: ");
    scanf("%d", &n);
```

```c
    for (int i = 0; i < n; i++) {
        scanf("%d", &val);
        insert(val);
    }

    printf("Inorder Traversal:\n");
    inorder(root);
}
```

OUTPUT: -

```
bharg@HP:~$ nano redblack.c
bharg@HP:~$ gcc redblack.c -o redblack
bharg@HP:~$ ./redblack
Enter number of nodes: 6
3
4
9
7
6
5
Inorder Traversal:
3(B) 4(B) 5(R) 6(B) 7(R) 9(B) bharg@HP:~$
```

AVL TREE: -

CODE: -

```
#include <stdio.h>
#include <stdlib.h>

struct Node {
   int key;
   struct Node *left;
   struct Node *right;
   int height;
};

int max(int a, int b) {
   return (a > b) ? a : b;
}

int height(struct Node *N) {
   if (N == NULL)
      return 0;
   return N->height;
}

struct Node* newNode(int key) {
   struct Node* node = (struct Node*)malloc(sizeof(struct Node));
   node->key = key;
   node->left = NULL;
   node->right = NULL;
   node->height = 1;
   return node;
}
```

```
struct Node *rightRotate(struct Node *y) {

    struct Node *x = y->left;
    struct Node *T2 = x->right;

    x->right = y;
    y->left = T2;

    y->height = max(height(y->left), height(y->right)) + 1;
    x->height = max(height(x->left), height(x->right)) + 1;

    return x;
}

struct Node *leftRotate(struct Node *x) {

    struct Node *y = x->right;
    struct Node *T2 = y->left;

    y->left = x;
    x->right = T2;

    x->height = max(height(x->left), height(x->right)) + 1;
    y->height = max(height(y->left), height(y->right)) + 1;

    return y;
}

int getBalance(struct Node *N) {
    if (N == NULL)
        return 0;
```

```
    return height(N->left) - height(N->right);
}

struct Node* insert(struct Node* node, int key) {

    if (node == NULL)
        return newNode(key);

    if (key < node->key)
        node->left = insert(node->left, key);

    else if (key > node->key)
        node->right = insert(node->right, key);

    else
        return node;

    node->height = 1 + max(height(node->left), height(node->right));

    int balance = getBalance(node);

    if (balance > 1 && key < node->left->key)
        return rightRotate(node);

    if (balance < -1 && key > node->right->key)
        return leftRotate(node);

    if (balance > 1 && key > node->left->key) {
        node->left = leftRotate(node->left);
        return rightRotate(node);
```

```
    }

    if (balance < -1 && key < node->right->key) {
        node->right = rightRotate(node->right);
        return leftRotate(node);
    }

    return node;
}

void inorder(struct Node *root) {

    if (root != NULL) {
        inorder(root->left);
        printf("%d ", root->key);
        inorder(root->right);
    }
}

int main() {

    struct Node *root = NULL;
    int n, value;

    printf("Enter number of nodes: ");
    scanf("%d", &n);

    printf("Enter values:\n");

    for(int i = 0; i < n; i++) {
        scanf("%d", &value);
```

```
        root = insert(root, value);
    }

    printf("\nInorder Traversal of AVL Tree:\n");
    inorder(root);

    printf("\n");

    return 0;
}
```

OUTPUT: -

```
bharg@HP:~$ nano AVL.c
bharg@HP:~$ gcc AVL.c -o avl
bharg@HP:~$ ./avl
Enter number of nodes: 6
Enter values:
7
6
2
8
1
5

Inorder Traversal of AVL Tree:
1 2 5 6 7 8
```